

Computational Replication Report:
Sobral, Mandal & Scheurer (2024),
Fractionalized Altermagnets (arXiv:2410.10949)

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1 Paper summary

The paper studies quantum-fluctuation-driven fractionalized phases near collinear altermagnetism. It identifies a non-coplanar “orbital altermagnet” (altermagnetic symmetry in spin-rotation-invariant observables), describes neighboring fractionalized spin liquids (Schwinger-boson / $SU(2) \rightarrow Z_2$ gauge theory), and — the subtitle result — shows that the doped fractionalized altermagnet has an electronic spectral function with *split Fermi surfaces that preserve spin-rotation symmetry: a spin-symmetric band splitting*.

2 Scope statement

The full self-consistent Schwinger-boson mean field, the gauge classification of the spin liquids, and the spinon–chargon convolution are deep many-body theory (out of scope). We implement the Appendix-C chargon spectral model (Eqs. C19–C21) exactly and reproduce the headline spin-symmetric splitting plus its contrast with ordinary altermagnetism.

3 Model (Eqs. C19–C21)

$$\begin{aligned}\epsilon_{\mathbf{k}} &= -t'(\cos k_x + \cos k_y) - \mu, \\ (g_{\mathbf{k}})_x + i(g_{\mathbf{k}})_y &= t(1 + e^{ik_x} + e^{i(k_x - k_y)} + e^{-ik_y}), \\ (g_{\mathbf{k}})_z &= -t(\cos k_x - \cos k_y) + H_0\alpha, \quad E_{\mathbf{k}}^{\pm} = \epsilon_{\mathbf{k}} \pm |g_{\mathbf{k}}|,\end{aligned}$$

with $\mathbf{g} \cdot \boldsymbol{\tau}$ acting in *sublattice* space. $t=1, t'=0.4, \mu=-0.3, H_0=0.5$.

4 Claims addressed

5 Results vs. paper

6 Per-claim what worked / what didn't

C1 (worked). $2|g_{\mathbf{k}}| > 0$ gives split Fermi surfaces; the d -wave part of g_z vanishes on the diagonals (node) and peaks on the axes — altermagnetic anisotropy. **C2 (worked — the headline).**

ID	Claim	Type	Testable?	Tes
C1	Split Fermi surfaces with d -wave-anisotropic $2 g_{\mathbf{k}} $	spectrum	yes	y
C2	Spin-symmetric: each split band spin-degenerate	symmetry	yes	y
C3	Contrast: ordinary altermagnet (form factor in spin) is spin-polarized	symmetry	yes	y
C4	Self-consistent SB mean field; spin-liquid classification; spinon-chargon convolution	many-body	no	out o

Quantity	Expected	This work	Verdict
Band splitting $2 g_{\mathbf{k}} $	> 0 (split FS)	$[0, 8.06]$	✓
d -wave part of g_z : axis vs diagonal	max vs node	2.0 vs 0.0	✓
Within-band spin splitting (fractionalized)	0 (spin-symmetric)	2.7×10^{-15}	✓
Spin polarization, ordinary altermagnet	~ 1 (spin-split)	1.000	✓

Because $\mathbf{g} \cdot \boldsymbol{\tau}$ lives in sublattice space, every band is a spin-degenerate doublet (within-band spin splitting $\sim 10^{-15}$): the splitting preserves spin-rotation symmetry. **C3 (worked)**. Placing the same d -wave form factor in *spin* space (ordinary altermagnet) gives spin polarization ≈ 1 — qualitatively different, confirming the distinction. **C4 (out of scope)**. The self-consistent SB mean field, the spin-liquid classification, and the full electron spectral function (spinon $\otimes$ chargon) are deep many-body theory.

7 Critique of the replication

Honest: we reproduce the paper’s *conceptual headline* (spin-symmetric band splitting) exactly from the App-C chargon equations, and cleanly contrast it with an ordinary altermagnet. But this is a BARE chargon dispersion — the physical electron spectral function (Fig. 3) requires convolution with the spinon Green’s function, which we did not compute (Open Q1); nor did we solve the self-consistent mean field selecting the fractionalized phase (Open Q2). C2’s near-zero within-band splitting is exact by the sublattice-vs-spin structure of the model, so it is a structural (not fitted) result. One subtlety was handled correctly: individual degenerate eigenvectors have arbitrary $\langle S_z \rangle$, so spin-rotation symmetry was verified via band DEGENERACY, not $\langle S_z \rangle$. LLM-judge: PARTIAL (coverage 5, agreement 6), adopted.

8 Verdict

PARTIAL — the headline spin-symmetric band splitting (split d -wave-anisotropic Fermi surfaces with preserved spin-rotation symmetry) and its contrast with ordinary altermagnetism are reproduced exactly from the App-C chargon model; the full Schwinger-boson mean field, spin-liquid classification, and spinon-chargon electron spectral function are out of scope.

Open Questions

See `report/open_questions.json`: Q1 spinon-chargon convolution; Q2 self-consistent phase selection; Q3 orbital-altermagnet $\rightarrow$ spin-liquid melting; Q4 experimental probe of spin-symmetric splitting; Q5 SR-invariant anomalous transport.